

MCV4U1 Exam Review - Calculus

LIMITS

1. The point $P(1, 4)$ lies on the parabola $y = x^2 + 3x$.
 - a) Find the slope of the tangent to the parabola at P .
 - b) Find an equation of the tangent to the parabola at P .

2. Let
$$f(x) = \begin{cases} 2x & \text{if } x \in (-\infty, -1] \\ x^2 & \text{if } x \in (-1, 1) \\ 2x - 1 & \text{if } x \in [1, \infty) \end{cases}$$
 - a) Find each limit, if it exists.
 - i) $\lim_{x \rightarrow -1^-} f(x)$
 - ii) $\lim_{x \rightarrow -1^+} f(x)$
 - iii) $\lim_{x \rightarrow -1} f(x)$
 - iv) $\lim_{x \rightarrow 1^-} f(x)$
 - v) $\lim_{x \rightarrow 1^+} f(x)$
 - vi) $\lim_{x \rightarrow 1} f(x)$
 - b) Where is f discontinuous?
 - c) How can we identify a discontinuity using limits?

3. Find each limit, if it exists.
 - a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 5x + 6}$
 - b) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 + 2x - 3}$
 - c) $\lim_{x \rightarrow 3} \frac{3x - 9}{2x^2 - 4x + 12}$
 - d) $\lim_{x \rightarrow 0} \frac{\sqrt{9 - x} - 3}{x}$
 - e) $\lim_{x \rightarrow -2} \frac{x^3 + 8}{x^2 - 4}$
 - f) $\lim_{x \rightarrow -4} \frac{\frac{4}{x} + 1}{x + 4}$

4. For each curve, find the slope of the tangent at the given point.
 - a) $y = 2x^2 - 3x + 4$ at $(1, 3)$
 - b) $y = \frac{x - 1}{x + 2}$ at $(-1, -2)$
 - c) $y = x^3 - 4$ at $(2, 4)$

5. For a fair start in any race, each competitor should be at the starting line or behind it, and should start from rest. The displacement, in metres, of a cyclist from the starting line, for the first 5 s of a race, is given by $s(t) = \frac{1}{2}t^2 + 2t$, where t is measured in seconds.
 - a) Find the average velocity for the cyclist over the first 5 s of the race.
 - b) Find the instantaneous velocity of the cyclist at $t = 5$.
 - c) Did the cyclist start the race fairly? Explain.

6. The population of a town for the next 15 years can be represented by $P(t) = -t^2 - 50t + 1200$.
 - a) Find the average rate of change of the population of the town for the first 10 years.
 - b) What is the instantaneous rate of change of the population at $t = 5$?
 - c) Describe what is happening to the number of people in the town.
 - d) What could cause this to occur in an average town?

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SOLUTIONS - LIMITS

1. a) 5 b) $y = 5x - 1$
2. a) i) -2 ii) 1 iii) does not exist iv) 1
v) 1 vi) 1
- b) at $x = -1$ c) A discontinuity exists when the two-sided limit does not exist.
3. a) -4 b) $\frac{3}{4}$ c) 0 d) $-\frac{1}{6}$ e) -3 f) $-\frac{1}{4}$
4. a) 1 b) 3 c) 12
5. a) 4.5 m/s b) 7 m/s c) No. When the race started, the cyclist was moving at 2 m/s.
6. a) -60 people/year b) -60 people/year
c) The population is decreasing.
d) Answers may vary. People may be moving out of the area faster than they are moving in, or dying faster than being born.

DERIVATIVE RULES

1. Use a first principles definition of the derivative to find the derivative of each function.
a) $f(x) = x^2 - 2x + 1$ b) $f(x) = \frac{1}{1-x}$
2. Use the derivative rules to find the derivative of each function.
a) $f(x) = 4 - 3x + 5x^2$ b) $g(x) = \sqrt[5]{x^3}$
c) $h(x) = \frac{1}{2x^3} + x - 3$ d) $g(x) = x^{-2}(2x^2 + 3x^3 - x^4)$
e) $f(x) = (2x^3 - 3x)(4 - x^2)$ f) $s(t) = \frac{1+t}{1+t^2}$
3. Find the slope of the tangent to each curve at the given point.
a) $f(x) = x^3 - 3x^2 + 4$, (2, 0) b) $f(x) = (x+2)(1-x^2)$, (2, -12)
c) $g(x) = \frac{x+3}{x-2}$, (1, -4) d) $h(x) = \frac{2x+1}{1-3x}$, (2, -1)
4. Find the second derivative of each function at $x = 4$.
a) $f(x) = 3x^2 + 2x + 1$ b) $y = \sqrt{x}(2x-1)$ c) $g(x) = \frac{x+1}{x-1}$
5. Let m and n represent the zeros of the parabola $f(x) = ax^2 + bx + c$, where a , b , and c are constants.
a) Find the zeros in terms of a , b , and c .
b) Show that if m and n are not equal, $f'(m) = -f'(n)$.
c) Find $f'(m)$ and $f'(n)$ if m is equal to n .
6. When charged particles, such as electrons, move through electric and magnetic fields, they follow curved paths. If there are no fields present, the particles experience no force and move in straight lines. An electron moves into a region of space with both electric and magnetic fields and follows the path $y = x^3 - 8x$ for $x > 0$. At some point along this path, the electron leaves the fields and moves in a straight line through the point (4, 0). At what point did the electron leave the fields?

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7. A company is marketing a new electronic organizer. The company claims the demand function for the product is $p(x) = \frac{30\,000 - x}{1000}$ and the cost of producing x organizers is $C(x) = 10\,000 + 2x + 0.001x^2$.
- Derive the revenue function.
 - Derive the marginal revenue function.
 - Derive the profit function.
 - Derive the marginal profit function.
 - Determine the marginal profit for $x = 10\,000$.

SOLUTIONS - DERIVATIVE RULES

- $2x - 2$
 - $(1 - x)^{-2}$
- $-3 + 10x$
 - $\frac{3}{5}x^{-\frac{2}{5}}$
 - $\frac{-3}{2x^4} + 1$
- $3 - 2x$
 - $-10x^4 + 33x^2 - 12$
- $\frac{1 - 2t - t^2}{(1 + t^2)^2}$
- 0
 - 19
 - 5
 - $\frac{1}{5}$
- 6
 - $\frac{25}{32}$
 - $\frac{4}{27}$
- $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 - $f'(m) = f'(n) = 0$
- (2, -8)
- $R(x) = 30x - 0.001x^2$
 - $R(x) = 30 - 0.002x$
 - $P(x) = 28x - 0.002x^2 - 10\,000$
 - $P(x) = 28 - 0.004x$
 - 12

CHAIN RULE

- Differentiate.
 - $y = (x^5 - 3x^4)^3$
 - $y = \sqrt[3]{x - 5}$
 - $y = (x^2 - 2)^5(3x - 4)$
 - $y = \frac{x^3 + 3}{\sqrt{x^3 - 5}}$
 - $y = \frac{x^3 + 3}{x^3 - 3}$
 - $y = \frac{2}{x^2 - 2x + 5}$
 - $y = \sqrt{x^5 + 3x}$
 - $y = \sqrt{(x^2 - 1)^3}$

SOLUTIONS - CHAIN RULE

- $3x^{11}(x - 3)^2(5x - 12)$
 - $\frac{1}{3(x - 5)^{\frac{2}{3}}}$
 - $(x^2 - 2)^4(33x^2 - 40x - 6)$
 - $\frac{3x^2(x^3 - 13)}{2(x^3 - 5)^{\frac{3}{2}}}$
 - $-\frac{18x^2}{(x^3 - 3)^2}$
 - $-\frac{4(x - 1)}{(x^2 - 2x + 5)^2}$
 - $\frac{5x^4 + 3}{2\sqrt{x^5 + 3x}}$
 - $3x\sqrt{x^2 - 1}$

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CURVE SKETCHING

1. Determine each limit.

a) $\lim_{x \rightarrow -3^-} \frac{x^2 + 9}{x + 3}$ b) $\lim_{x \rightarrow -3^+} \frac{x^2 + 9}{x + 3}$ c) $\lim_{x \rightarrow 2^-} \frac{3 - x^2}{x^2 - 5x + 6}$ d) $\lim_{x \rightarrow 2^+} \frac{3 - x^2}{x^2 - 5x + 6}$

2. Determine the equations all the asymptotes of each curve.

a) $y = \frac{9x}{x + 6}$ b) $y = \frac{2x^2}{x - 3}$ c) $y = \frac{3x^2}{8 - x^3}$

3. Determine the critical numbers of each function and determine whether the critical numbers correspond to local maximum points or local minimum points.

a) $y = x^2 - 3x - 4$ b) $y = 2x^3 - 36x$ c) $y = \frac{x}{x^2 + 1}$

4. Determine the critical numbers and find the absolute maximum and minimum values of the function $y = 8x^3 + x^4$ for $x \in [-8, 2]$.

5. a) Determine the intervals on which the curve $y = 6 - 3x^2 + x^3$ is concave upward and the intervals on which it is concave downward.
b) Determine if there are any points of inflection.

6. For the curve $y = x^4 - 4x^3$, determine:

- a) the intervals of increase or decrease b) the local maximum and minimum values
c) the intervals of concavity d) any points of inflection

7. Sketch the function $y = \frac{2x^2}{x^2 - 1}$. Determine enough information to sketch an accurate graph. You should include intercepts, asymptotes, extrema, points of inflection, and any other significant details, justifying your results with analytic methods.

8. The pages in a textbook must have top and bottom margins 4 cm deep, and left and right margins 2 cm wide. The rectangular text area in the centre of the page must be 800 cm². Find the dimensions of the page that will minimize the page area.

9. A manufacturer can produce x washing machines at a cost, in dollars, modelled by $C(x) = 4000 + 150x + 0.5x^2$. The demand function for the washing machines is $p(x) = 600 - 3x$. What production level will maximize profit?

SOLUTIONS – CURVE SKETCHING

1. a) $-\infty$ b) ∞ c) $-\infty$ d) ∞

2. a) $x = -6$, $y = 9$

b) $x = 3$, $y = 2x + 6$

c) $x = 2$, $y = 0$

3. a) critical number: $x = \frac{3}{2}$;

minimum: $-\frac{25}{4}$ at $x = \frac{3}{2}$

b) critical numbers: $x = -\sqrt{6}, \sqrt{6}$;

maximum: $24\sqrt{6}$ at $x = -\sqrt{6}$;

minimum: $-24\sqrt{6}$ at $x = \sqrt{6}$

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c) critical numbers: $x = -1, 1$;

minimum: $-\frac{1}{2}$ at $x = -1$; maximum: $\frac{1}{2}$ at $x = 1$

4. critical numbers: $x = 0, -6$;

minimum: -432 ; maximum: 80

5. a) concave upward: $(1, \infty)$;

concave downward: $(-\infty, 1)$ b) $(1, 4)$

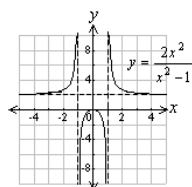
6. a) increase: $(3, \infty)$; decrease: $(-\infty, 3)$

b) minimum: -27

c) concave upward: $(-\infty, 0), (2, \infty)$;

concave downward: $(0, 2)$

d) $(0, 0), (2, -16)$



7.

x - and y -intercepts: 0 ;

asymptotes: $x = 1$ and -1 , $y = 2$;

maximum point: $(0, 0)$

8. 24 cm by 48 cm

9. 64

DERIVATIVES OF EXPONENTIAL AND LOGARITHMIC FUNCTIONS

1. Differentiate with respect to x .

a) $y = e^{5x} + a^{5x}$

b) $y = \frac{\ln x}{x^5}$

c) $y = \ln \sqrt[3]{3x^2 - 4}$

2. Determine an equation of the tangent to $y = \frac{\ln x}{x}$ at the point $(1, 0)$.

2. Determine the local extrema, inflection points, and all asymptotes of the function $y = xe^{-x}$. Use this information to make a clearly labelled sketch of the function.

SOLUTIONS – DERIVATIVES OF EXPONENTIAL AND LOGARITHMIC FUNCTIONS

1. a) $5e^{5x} + 5a^{5x} \ln a$ b) $\frac{1 - 5 \ln x}{x^6}$ c) $\frac{2x}{3x^2 - 4}$

2. $y = x - 1$

3. maximum: $(1, e^{-1})$;

point of inflection: $(2, 2e^{-2})$;

asymptote: $y = 0$

